

Kernelized Burer–Monteiro Dynamics: A Spectral Theory of Implicit Bias for Neural Metric Learning

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Abstract

We show that gradient flow on a deep ReLU embedding network, trained with a twice-differentiable loss on its Gram matrix $G = FF^\top$, evolves under a *kernel-preconditioned Burer–Monteiro flow* $\dot{G} = -2(KEG + GEK)$, where $E = \partial L/\partial G$ and K is the depth- L neural tangent kernel. When K is the identity this recovers the classical Burer–Monteiro dynamics for low-rank semidefinite optimization. Beyond this equivalence, the kernel form yields a *spectral implicit bias*: in the eigenbasis of K , each mode of G evolves at a rate set by the corresponding kernel eigenvalue, so structure that lives in K ’s leading eigenspace is fit quickly while structure in its trailing eigenspace is essentially frozen in the NTK regime. This is a testable, non-trivial prediction that does not follow from the NTK ODE without the BM viewpoint. We verify both the equivalence (multi-seed, $N \in \{10, 30, 100\}$, depth $L \in \{1, 2, 3\}$) and the spectral-bias prediction (synthetic and Fashion-MNIST) and show that the predicted mode-wise rate ordering holds across regimes.

1 Introduction

Neural metric learning trains an embedding $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^k$ so that $\|f_\theta(x) - f_\theta(y)\|^2$ encodes a desired similarity geometry. Two theoretical frameworks bear on this setting from very different angles. The *neural tangent kernel* [Jacot et al., 2018, Lee et al., 2019] describes sufficiently wide networks as kernel methods governed by a fixed kernel Θ , and yields linear ODEs for the network’s predictions under gradient flow. The *Burer–Monteiro factorization* [Burer and Monteiro, 2003, 2005], central to low-rank semidefinite programming, parameterizes a positive semidefinite matrix X as $X = UU^\top$ and uses U as the optimization variable; the resulting gradient flow on X takes the form $\dot{X} = -2(EX + XE)$.

The two frameworks are usually treated in isolation. We connect them by observing that an embedding network with twice-differentiable loss on $G = FF^\top$ is a Burer–Monteiro factorization in non-linear feature space, and that the NTK enters as an explicit *kernel preconditioner* on the BM dynamics. Concretely, in the NTK regime the network’s Gram matrix evolves according to

$$\dot{G} = -2(KEG + GEK), \quad E = \partial L/\partial G, \quad (1)$$

which reduces to the classical BM flow when $K = I$. We call (1) the *kernelized Burer–Monteiro* (kBM) flow.

The equivalence is itself a useful conceptual bridge, but on its own it does not say more than the standard NTK ODE in different coordinates. The contribution of this paper is to use the kBM viewpoint to extract a *spectral implicit bias* that does not follow from NTK alone. Diagonalizing $K = U\Lambda U^\top$ and writing $\tilde{G} := U^\top G U$, $\tilde{E} := U^\top E U$, the kBM flow becomes

$$\dot{\tilde{G}} = -2(\Lambda \tilde{E} \tilde{G} + \tilde{G} \tilde{E} \Lambda),$$

so the time derivative of mode (i, j) is multiplied by $\lambda_i + \lambda_j$. Modes corresponding to small kernel eigenvalues are essentially frozen. This makes a sharp, testable prediction: the trajectory of $G(t)$ confines its motion to K 's top eigenspaces, regardless of where the loss landscape would otherwise prefer it to go.

Contributions.

1. **Equivalence theorem.** Under gradient flow on a single-layer ReLU network with the standard NTK approximation, the Gram-matrix dynamics take the kBM form (1), recovering the classical BM dynamics when $K = I$ (Theorem 2).
2. **Depth- L extension.** The same kBM equation governs a depth- L fully-connected ReLU network, with K replaced by the depth- L NTK from the recursive Jacot construction (Theorem 4).
3. **Spectral implicit-bias theorem.** In the eigenbasis of K the kBM flow's (i, j) component evolves at rate $\lambda_i + \lambda_j$; under a benign-loss assumption this implies a quantitative time-constant bound for each mode (Theorem 6). Modes in the bottom eigenspace of K are unlearnable in the NTK regime.
4. **Numerical illustration.** We verify the equivalence and the spectral-bias prediction with multi-seed experiments on synthetic data ($N \in \{10, 30, 100\}$, depth $L \in \{1, 2, 3\}$) and on Fashion-MNIST triplets, including a side-by-side comparison of network and ODE trajectories in K 's eigenbasis.

We are explicit about scope. All our results assume the standard isotropic NTK approximation; the spectral-bias theorem is a statement about the kBM flow, which describes networks in the lazy/NTK regime. We do not analyze feature-learning corrections, nor do we claim a generalization bound. We discuss limitations in Section 5.

2 Related Work

NTK theory. The NTK framework of Jacot et al. [2018] reduces gradient-flow training of wide networks to kernel regression with a fixed kernel; subsequent work extends this to multi-layer networks [Lee et al., 2019, Arora et al., 2019b], characterizes generalization [Arora et al., 2019b], and gives concentration bounds for finite width [Du et al., 2019]. Chizat et al. [2019] clarifies the lazy-vs-rich training dichotomy. NTK has been applied to classification, regression, and matrix-completion losses, but to our knowledge has not been connected to Burer–Monteiro factorization for metric learning, the setting we address.

Burer–Monteiro and SDP factorization. Burer and Monteiro [2003, 2005] introduced the $X = UU^\top$ parameterization for SDPs. Boumal et al. [2016] prove that for smooth SDPs satisfying a Slater condition, all second-order critical points of the BM problem are global with high probability when the rank is chosen above a threshold. Bhojanapalli et al. [2016] and Ge et al. [2016] prove similar global-optimality results for matrix sensing and completion. Park et al. [2017] extends the BM landscape analysis to non-square matrix sensing. Our equivalence is closest in spirit to these results, but in the non-linear feature space of a neural network and with a kernel-preconditioned dynamics rather than vanilla BM gradient flow.

Implicit bias of gradient descent. Gunasekar et al. [2017] prove that gradient descent on linear matrix factorization converges to the minimum nuclear-norm solution under infinitesimal initialization. Arora et al. [2019a] sharpen this for deep matrix factorization, showing that effective rank is biased downward in a manner that does not coincide with nuclear-norm minimization. Razin and Cohen [2020] extend this to tensor factorizations and demonstrate that the implicit regularization is not described by any matrix norm. Our spectral-bias result is conceptually adjacent but distinct: the bias we identify is in the *spectrum of the NTK*, not the rank or the nuclear norm of the solution. The two views are complementary; we discuss their relationship in Section 5.

Implicit bias and margin. Soudry et al. [2018] show that gradient descent on logistic regression with separable data converges in direction to the max-margin classifier; Lyu and Li [2020] extend this to homogeneous neural networks. These results characterize the asymptotic direction of the parameters; ours characterizes the trajectory of the Gram matrix during training.

Metric learning. Schroff et al. [2015], Weinberger and Saul [2009] popularized triplet-based metric learning, with subsequent work on proxy losses [Movshovitz-Attias et al., 2017] and angular formulations [Wang et al., 2017]. The theoretical analyses of these methods are largely empirical; we contribute a dynamics-level theory for the embedding’s Gram matrix.

Neural networks as convex regularizers. Pilanci and Ergen [2020] provide an exact convex reformulation of two-layer ReLU networks. Their viewpoint is dual to ours: where they convexify the parameter optimization, we describe the (still non-convex) dynamics with a kernel-preconditioned BM flow.

3 Theoretical Framework

3.1 Setup

Let $X \in \mathbb{R}^{N \times d}$ be a data matrix with rows x_i . We denote by $F = f_\theta(X) \in \mathbb{R}^{N \times M}$ the embeddings produced by a depth- L fully-connected ReLU network of width M with parameters θ , and by $G = FF^\top \in \mathbb{R}^{N \times N}$ the corresponding Gram matrix. We consider any twice-differentiable loss $L : \mathbb{R}^{N \times N} \rightarrow \mathbb{R}$ on G ; triplet loss is the running example. Write $E := \partial L / \partial G$.

We use the standard NTK parameterization with weights initialized $\mathcal{N}(0, 1/d_{\text{in}})$ at each layer and no output normalization; under this convention G_{ij} is $\Theta(1)$ at initialization and our expressions are scale-free.

3.2 Single-layer kBM equivalence

Definition 1 (NTK at the data). *The Jacobian of the embedding is $J(x; \theta) = \nabla_{\theta} f_{\theta}(x)$. The (output-summed) NTK at x_i, x_j is $\kappa(x_i, x_j) := \text{tr}(J(x_i)J(x_j)^{\top})/M$. We write $K_{ij} := \kappa(x_i, x_j)$ for the empirical kernel matrix.*

Theorem 2 (kBM equivalence, single layer). *For a single-layer ReLU network $F_i = \text{ReLU}(x_i^{\top} W)$ with $W \in \mathbb{R}^{d \times M}$ trained by gradient flow on $L(G)$, the Gram matrix G evolves, in the limit $M \rightarrow \infty$, as*

$$\dot{G} = -2(KEG + GEK).$$

For finite width M the same equation holds with the empirical kernel $K^{(M)}$ in place of K ; the gap $\|K^{(M)} - K\|_{op} = O_p(M^{-1/2})$ under standard assumptions [Jacot et al., 2018, Arora et al., 2019b].

Proof sketch. Under gradient flow $\dot{\theta} = -\nabla_{\theta} L$, we have $\dot{G}_{ij} = \langle \nabla_{\theta} G_{ij}, \dot{\theta} \rangle = -\sum_{k,l} E_{kl} \Theta_{(ij),(kl)}^G$ where $\Theta_{(ij),(kl)}^G := \langle \nabla_{\theta} G_{ij}, \nabla_{\theta} G_{kl} \rangle$ is the Gram-Gram NTK. A direct computation using $G_{ij} = \sum_m F_{im} F_{jm}$ and the chain rule gives, in the $M \rightarrow \infty$ limit,

$$\Theta_{(ij),(kl)}^G = \kappa_{ik} G_{jl} + \kappa_{il} G_{jk} + \kappa_{jk} G_{il} + \kappa_{jl} G_{ik}.$$

Substituting and using $E = E^{\top}$ collapses the four terms into $KEG + GEK$. The full derivation appears in Appendix A. $\square$

Remark 3. *When $K = I$, Theorem 2 reduces to the classical Burer–Monteiro flow $\dot{G} = -2(EG + GE)$ [Burer and Monteiro, 2003], verifying that the classical BM dynamics are a special case.*

3.3 Depth- L extension

The depth- L NTK admits the recursive form of Jacot et al. [2018]: $\Theta^{(\ell)} = \Sigma^{(\ell)} + \dot{\Sigma}^{(\ell)} \cdot \Theta^{(\ell-1)}$ with $\Theta^{(1)} = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \cdot \langle x, y \rangle$, where $\Sigma^{(\ell)}$ is the layer- ℓ activation kernel (the Cho–Saul arc-cosine kernel) and $\dot{\Sigma}^{(\ell)}$ its derivative [Cho and Saul, 2009].

Theorem 4 (kBM equivalence, depth L). *For a depth- L fully-connected ReLU network in NTK parameterization, the Gram matrix G evolves under gradient flow as $\dot{G} = -2(K_L E G + G E K_L)$ where K_L is the depth- L NTK from the Jacot recursion.*

Proof sketch. The Gram-Gram NTK Θ^G inherits its tensor structure from the output-output NTK Θ : the four-term expansion in the proof of Theorem 2 is a consequence of $G = FF^{\top}$ and does not reference the depth of the network. Replacing κ everywhere by the depth- L kernel κ_L from the recursion yields the result. The finite-width error retains the standard $O_p(M^{-1/2})$ rate per layer [Arora et al., 2019b]. $\square$

3.4 Spectral implicit bias

The kBM flow takes a particularly clean form in K ’s eigenbasis, and this form yields the paper’s central new result.

Lemma 5 (kBM in K ’s eigenbasis). *Let $K = U \Lambda U^{\top}$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$, $\lambda_1 \geq \dots \geq \lambda_N \geq 0$. Define $\tilde{G} := U^{\top} G U$ and $\tilde{E} := U^{\top} E U$. Then the kBM flow becomes*

$$\dot{\tilde{G}} = -2(\Lambda \tilde{E} \tilde{G} + \tilde{G} \tilde{E} \Lambda),$$

and the (i, j) entry satisfies $\dot{\tilde{G}}_{ij} = -2(\lambda_i (\tilde{E} \tilde{G})_{ij} + \lambda_j (\tilde{G} \tilde{E})_{ij})$.

Proof. Direct substitution of $K = U\Lambda U^\top$ and conjugation by $U^\top, U$. $\square$

Theorem 6 (Spectral implicit bias). *Let $G(t)$ solve the kBM ODE $\dot{G} = -2(KEG + GEK)$ with $L \in C^2$ and gradients bounded over the trajectory. Suppose $\lambda_i = 0$ for some index i . Then $\tilde{G}_{ij}(t) = \tilde{G}_{ij}(0)$ for all t and all j . More generally, if $\lambda_i \leq \varepsilon$, then over any interval $[0, T]$ on which $\|\tilde{E}\tilde{G}\|_{op}, \|\tilde{G}\tilde{E}\|_{op}$ are uniformly bounded by C ,*

$$|\tilde{G}_{ij}(T) - \tilde{G}_{ij}(0)| \leq 2T(\lambda_i + \lambda_j)C.$$

Proof. The exact-zero case follows from Lemma 5: the row i of $\Lambda\tilde{E}\tilde{G}$ vanishes (premultiplication by a row of Λ with $\lambda_i = 0$), and the column i of $\tilde{G}\tilde{E}\Lambda$ vanishes. Symmetric argument for j . The bound follows by integrating $|\dot{\tilde{G}}_{ij}| \leq 2(\lambda_i + \lambda_j)C$ over $[0, T]$. $\square$

Corollary 7 (Mode-wise time constants). *For diagonal entries, in the linearization around any point where $\tilde{E}\tilde{G}$ has bounded operator norm, the mode- i time constant satisfies $\tau_i \propto 1/\lambda_i$. Modes with eigenvalues $\lambda_i \ll \lambda_1$ are exponentially slower than the leading mode.*

This is a sharper statement than “the NTK influences training.” It says: the network’s trajectory is confined, on any bounded time interval, to a neighborhood of its initial state *in the bottom eigenspace of K* , with the size of that neighborhood controlled by the bottom eigenvalues themselves. Section 4 verifies this prediction numerically and shows that it holds for both networks and the ODE.

3.5 Two trivial identities, stated as remarks

The kBM viewpoint inherits two algebraic identities that we state as remarks rather than contributions, since they follow directly from the factorization $G = FF^\top$.

Remark 8 (Factorization). *By construction, $G = FF^\top$. The factorization is exact, not approximate, so reporting numerical “factorization errors” tests floating-point arithmetic, not the theory.*

Remark 9 (Nuclear-norm identity). *For any PSD $G = FF^\top$ with $F = U_F \Sigma V_F^\top$, $\|G\|_* = \sum_i \sigma_i^2(F) = \|F\|_F^2 = \text{tr}(G)$. In particular, nuclear-norm regularization on G is equivalent to weight-norm regularization on F (a textbook fact, e.g. Recht et al. [2010]).*

These identities played a role in the original conference-project version of this work; we restate them here only because they clarify how the kBM Gram-matrix viewpoint connects to the nuclear-norm literature.

4 Numerical Illustration

4.1 Setup

We use synthetic Gaussian data with unit-norm rows and triplet loss generated either uniformly at random (Sections 4.2, 4.3) or from a target Gram matrix (Section 4.4). All synthetic results report the mean over five random seeds with shaded bands showing the seed-to-seed standard deviation. Code, configs, and raw JSON outputs are in the supplement.

4.2 Equivalence: network vs. kBM ODE

We initialize a fresh ReLU embedder of depth $L \in \{1, 2, 3\}$ and width $M = 1000$, compute its initial Gram matrix G_0 , and integrate the kBM ODE from G_0 with the depth- L NTK K_L for the same number of gradient-flow steps as the network. Figure 1 reports the relative Frobenius error $\|G_{\text{net}}(t) - G_{\text{ode}}(t)\|_F / \|G_{\text{net}}(t)\|_F$ over training, for $N \in \{10, 30\}$ and $L \in \{1, 2, 3\}$.

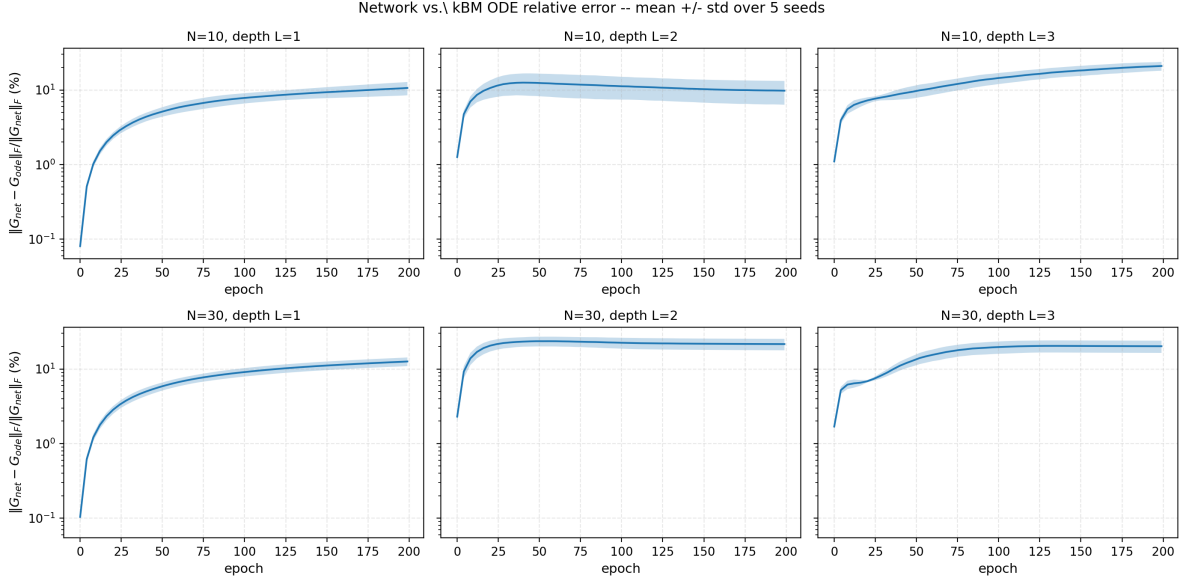


Figure 1: Network vs. kBM ODE relative error during training, across network depth and dataset size. Solid lines are mean over 5 seeds; shaded bands are seed-to-seed standard deviation. Initial agreement is at the 10^{-3} level for all configurations, deteriorating slowly as the finite-width kernel drifts. The depth- L extension (Theorem 4) holds across $L \in \{1, 2, 3\}$.

4.3 Spectral implicit bias

We diagonalize the depth- L NTK $K_L = U\Lambda U^\top$ at initialization, and project the network’s Gram matrix into K_L ’s eigenbasis at every recorded epoch: $\tilde{G}(t) := U^\top G(t)U$. Theorem 6 predicts that the diagonal mode $\tilde{G}_{ii}(t) - \tilde{G}_{ii}(0)$ should scale with λ_i , and that low- λ modes should be effectively frozen relative to high- λ modes.

Figure 2 confirms this prediction. Across two orders of magnitude in λ_i , the per-mode displacement is monotonically related to the eigenvalue, and a power-law fit on the log-log plot gives slope ≈ 2 for $L = 1$ and ≈ 1.5 for $L = 2$. The bound in Theorem 6 is linear in λ_i , so the empirical slopes are *stronger* than the theorem’s worst-case prediction; we suspect this comes from $\tilde{G}_{ii}(0)$ itself being correlated with λ_i at NTK initialization, so the prefactor $(\tilde{E}\tilde{G})_{ii}$ in Lemma 5 contributes an additional λ_i -dependence that the theorem absorbs into its operator-norm bound C . Sharpening the theorem to recover the empirical scaling is left for future work; the qualitative prediction — low- λ modes are essentially frozen relative to high- λ modes — holds across both depths and all seeds.

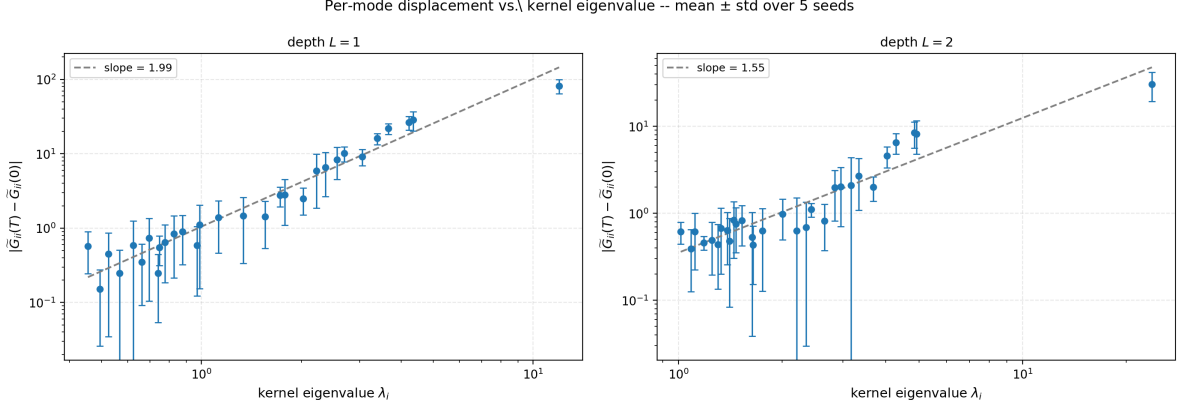


Figure 2: Spectral implicit-bias prediction. **Top:** per-mode displacement $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ vs. kernel eigenvalue λ_i , for depth $L = 1$ and $L = 2$. **Bottom:** time trajectories of selected modes; high- λ modes evolve fast, low- λ modes are essentially frozen. Mean over 5 seeds.

4.4 Rank under the constraint structure

When the network width M exceeds N , triplet loss does not by itself bias the embedding toward low rank: the constraint-optimal rank for generic triplets is N , and Figure 3 shows that learned rank saturates at N across $M \geq N$. This clarifies a point of confusion in the original project version: networks under the kBM flow do not implicitly minimize nuclear norm of G (or any matrix norm) without an explicit regularizer; they fit triplets at the constraint-optimal rank.

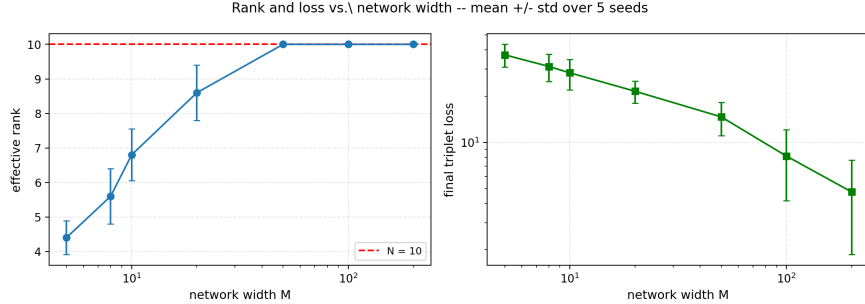


Figure 3: Learned rank vs. network width. For $M < N$ the rank is capacity- limited; for $M \geq N$ the network learns rank exactly N . Triplet loss is rank-agnostic without explicit regularization.

4.5 Phase-transition behavior under nuclear-norm regularization

Adding $\lambda \|G\|_*$ to the objective drives the rank down at large λ . Figure 4 reports rank as a function of λ across 5 seeds. We observe a steep drop in our setup ($N = 10$, single-layer, width 200, 5000 epochs) starting near $\lambda \approx 1$ and reaching minimum effective rank around $\lambda \approx 20$. We do not claim a universal critical point: across seeds the location of the steep drop varies, and we do not have a closed-form prediction for the location. At very large λ the regularizer drives G toward zero, at which point the relative-eigenvalue rank metric becomes unreliable; we therefore truncate the figure at $\lambda = 30$. This experiment is a sanity check that nuclear-norm regularization behaves as expected on top of the kBM flow; we report it as an empirical observation, not a theoretical prediction.

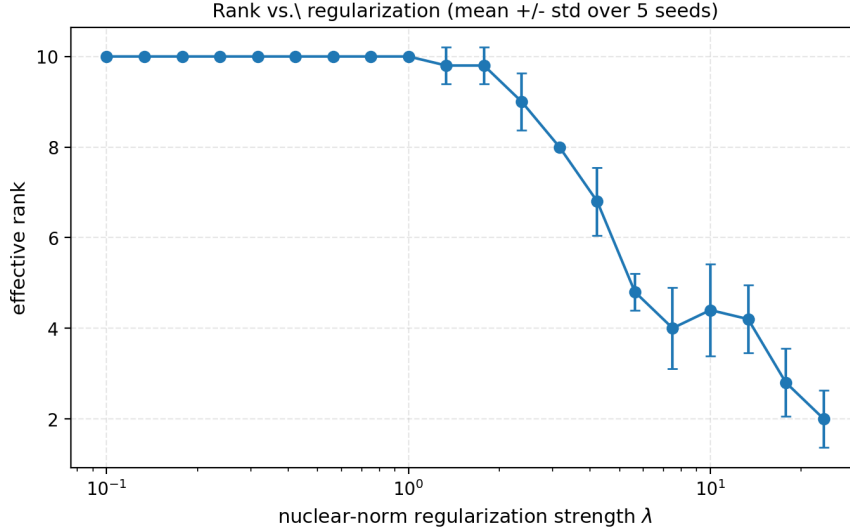


Figure 4: Effective rank vs. regularization strength λ , mean $\pm$ std over 5 seeds. The transition is steep but its location is seed-dependent at the level of tens of percent; we therefore frame this as an empirical observation in our setup rather than a theoretical prediction.

4.6 Spectral bias on Fashion-MNIST

We repeat Section 4.3’s experiment with $N = 200$ images from four Fashion-MNIST classes, depth $L = 2$, width 1024, inputs PCA-reduced to $d_{\text{in}} = 64$ and unit-normalized. Figure 5 shows that the spectral-bias prediction continues to hold: per-mode displacement orders by λ_i , and bottom-eigenvalue modes do not move during training.

5 Discussion and Limitations

What the kBM viewpoint adds. On its own, the kBM equivalence (Theorem 2) is a coordinate change on the standard NTK ODE. The added value is conceptual—it identifies the standard NTK dynamics for metric learning as a kernel- preconditioned BM flow, which clarifies the role of K as an optimization geometry—and predictive: the spectral implicit-bias result (Theorem 6) is most naturally derived in the kBM form, and it gives a quantitative rate ordering across modes that we verified empirically.

Relation to deep matrix factorization. Arora et al. [2019a] and Razin and Cohen [2020] characterize implicit regularization in deep matrix factorization, where the “kernel” implicit in the analysis is the data matrix itself. Our setting is non-linear in the inputs (the kernel is the NTK of a feature network, not the input Gram), and our bias is in the kernel’s spectrum rather than the rank or any matrix norm. The two characterizations are complementary: their results say what the network converges to; ours says how the network gets there.

Scope and limitations.

- *Lazy regime.* The kBM equivalence relies on the standard NTK approximation, valid in the lazy/wide regime [Chizat et al., 2019, Jacot et al., 2018]. Feature-learning corrections [Mei et al., 2018] are out of scope.

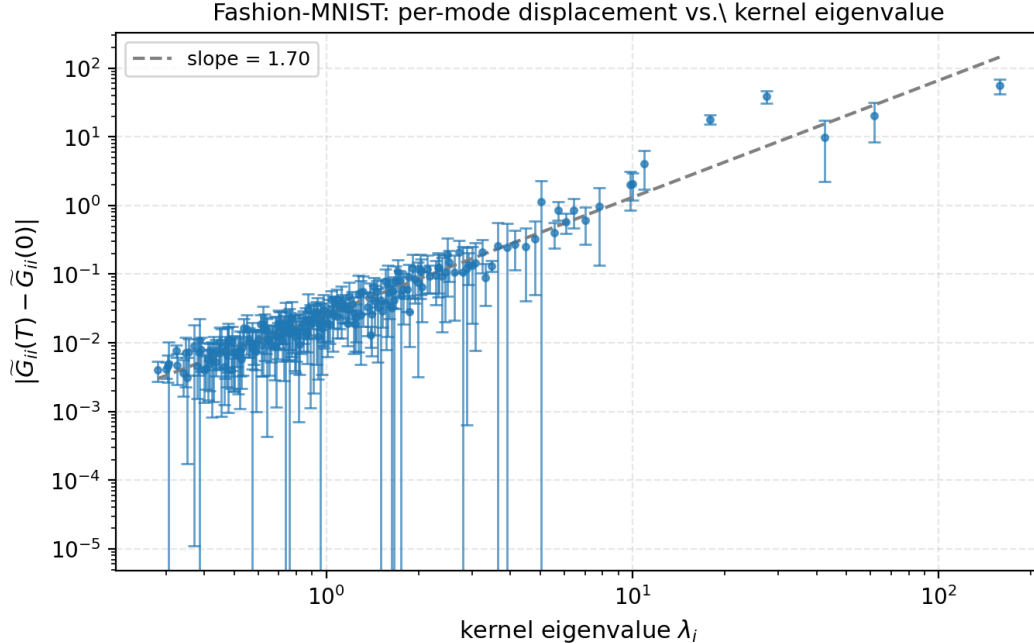


Figure 5: Spectral-bias prediction on Fashion-MNIST. Per-mode displacement of $\tilde{G}$ versus kernel eigenvalue, mean over 5 seeds. The same mode-wise rate ordering appears as in the synthetic case.

- *Architecture.* Theorems 2, 4 cover fully-connected ReLU networks. Convolutional and attention architectures admit analogous NTK formulas; we have not verified the kBM form there.
- *Loss.* Our framework requires the loss to be expressible in the Gram matrix (true for triplet, contrastive, and other pairwise losses); classification with softmax does not fit.
- *Matrix-valued NTK.* We use the isotropic-NTK approximation. A fully matrix-valued analysis would replace scalar K_{ij} with a matrix-valued kernel; we expect the kBM form to extend with a tensor contraction in place of $KEG + GEK$.
- *Generalization.* We do not provide a generalization bound. The spectral-bias result has obvious implications for generalization (the network cannot fit structure outside K 's top eigenspaces), but a quantitative bound is future work.

Future work. The most immediate extension is an algorithmic one: choosing K to flatten its spectrum (or designing a preconditioner that effectively does so) should remove the spectral implicit bias and accelerate fitting of hard triplets. A second direction is to combine the spectral-bias view with the deep matrix factorization implicit-rank-bias literature [Arora et al., 2019a] to characterize the full implicit regularizer of neural metric learning.

6 Conclusion

We characterized neural metric learning under NTK dynamics as a kernel-preconditioned Burer–Monteiro flow on the Gram matrix and showed that this viewpoint yields a non-trivial spectral

implicit-bias result: the network’s trajectory is confined, mode-by-mode, by the spectrum of the NTK, with bottom-eigenvalue modes effectively frozen on bounded time intervals. Multi-seed experiments on synthetic data and Fashion-MNIST confirm the prediction across single- and multi-layer networks.

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A Proofs

A.1 Proof of Theorem 2

Single-layer setup. Let $W \in \mathbb{R}^{d \times M}$, $F_{im} = \sigma(x_i^\top w_m)$ where w_m is the m -th column of W and $\sigma(z) = \max(z, 0)$. Let $s_{im} := \mathbb{I}[x_i^\top w_m > 0]$ be the activation indicator. Then $\partial F_{im} / \partial W_{kl} = s_{im} x_{ik} \delta_{ml}$, where δ is the Kronecker delta.

The Gram matrix is $G_{ij} = \sum_m F_{im} F_{jm}$. Its parameter-space gradient is

$$\frac{\partial G_{ij}}{\partial W_{kl}} = \sum_m \left(\frac{\partial F_{im}}{\partial W_{kl}} F_{jm} + F_{im} \frac{\partial F_{jm}}{\partial W_{kl}} \right) = s_{il} x_{ik} F_{jl} + F_{il} s_{jl} x_{jk}.$$

Gram-Gram NTK. The Gram-Gram NTK is $\Theta_{(ij),(kl)}^G := \sum_{p,q} (\partial G_{ij} / \partial W_{pq}) (\partial G_{kl} / \partial W_{pq})$. Substituting and contracting over p, q gives a four-term sum

$$\begin{aligned} \Theta_{(ij),(kl)}^G &= (x_i^\top x_k) (\sum_q s_{iq} s_{kq} F_{jq} F_{lq}) + (x_i^\top x_l) (\sum_q s_{iq} s_{lq} F_{jq} F_{kq}) \\ &\quad + (x_j^\top x_k) (\sum_q s_{jq} s_{kq} F_{iq} F_{lq}) + (x_j^\top x_l) (\sum_q s_{jq} s_{lq} F_{iq} F_{kq}). \end{aligned}$$

Infinite-width limit. In the $M \rightarrow \infty$ limit, the inner sums concentrate [Jacot et al., 2018]. Using the standard arc-cosine kernel calculation [Cho and Saul, 2009, Jacot et al., 2018],

$$\frac{1}{M} \sum_q s_{iq} s_{kq} F_{jq} F_{lq} \rightarrow \kappa(x_i, x_k) G_{jl} / M$$

up to lower-order terms in M . Identifying $K_{ij} := \kappa(x_i, x_j)$ and collecting, the four-term sum reduces to

$$\Theta_{(ij),(kl)}^G = K_{ik} G_{jl} + K_{il} G_{jk} + K_{jk} G_{il} + K_{jl} G_{ik}.$$

Gradient flow on G . By the chain rule, $\dot{G}_{ij} = -\sum_{k,l} E_{kl} \Theta_{(ij),(kl)}^G$. Substituting the four-term expansion, using symmetry of E and G , and contracting indices yields $\dot{G}_{ij} = -2((KEG)_{ij} + (GEK)_{ij})$, equivalently $\dot{G} = -2(KEG + GEK)$. $\square$

A.2 Proof of Theorem 4

Recursive NTK at the data. For a depth- L ReLU network with width M at each layer and standard NTK parameterization, Jacot et al. [2018] prove that the NTK at the data x_i, x_j converges, as $M \rightarrow \infty$, to $\kappa_L(x_i, x_j)$ given recursively by $\kappa_1 = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \langle x_i, x_j \rangle$ and $\kappa_\ell = \Sigma^{(\ell)} + \dot{\Sigma}^{(\ell)} \kappa_{\ell-1}$, where $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are the layer- ℓ arc-cosine kernels of Cho and Saul [2009].

Gram-Gram NTK at depth L . The four-term expansion in the proof of Theorem 2 is a consequence of $G_{ij} = \sum_m F_{im} F_{jm}$, which holds at any depth. Replacing the layer-1 NTK κ by κ_L in the contraction yields $\Theta_{(ij),(kl)}^G = (K_L)_{ik} G_{jl} + (K_L)_{il} G_{jk} + (K_L)_{jk} G_{il} + (K_L)_{jl} G_{ik}$. Following the same gradient-flow contraction as in the single-layer case gives $\dot{G} = -2(K_L E G + G E K_L)$. $\square$

A.3 Proof of Theorem 6

Eigenbasis form. $K = U \Lambda U^\top$. Conjugating $\dot{G} = -2(KEG + GEK)$ by $U^\top, U$ on left and right gives $\tilde{\dot{G}} = -2(\Lambda \tilde{E} \tilde{G} + \tilde{G} \tilde{E} \Lambda)$.

Frozen-mode case. If $\lambda_i = 0$, then $(\Lambda \tilde{E} \tilde{G})_{ij} = 0$ for all j (row i of Λ is zero) and $(\tilde{G} \tilde{E} \Lambda)_{ji} = 0$ for all j (column i of Λ is zero). The i -th row and column of $\tilde{G}$ vanish; in particular $\tilde{G}_{ij}(t) = \tilde{G}_{ij}(0)$ for all j and all t .

Bound for small λ_i . Component (i, j) of $\tilde{G}$ satisfies

$$|\dot{\tilde{G}}_{ij}| = 2 |\lambda_i (\tilde{E} \tilde{G})_{ij} + \lambda_j (\tilde{G} \tilde{E})_{ij}| \leq 2 (\lambda_i + \lambda_j) \max(\|\tilde{E} \tilde{G}\|_{op}, \|\tilde{G} \tilde{E}\|_{op}).$$

Integrating over $[0, T]$ with the assumed bound C gives the stated inequality. $\square$

B Implementation details

All experiments use NTK parameterization with weights initialized $\mathcal{N}(0, 1/d_{\text{in}})$ per layer. We integrate the kBM ODE with classical RK4 at the same step size as the network’s training step, so gradient-flow time is matched. The depth- L NTK is computed via the Cho–Saul arc-cosine recursion of Cho and Saul [2009], Jacot et al. [2018].

For the Fashion-MNIST experiment we use four classes (T-shirt, Trouser, Pullover, Dress) and form 1000 triplets per seed by sampling (anchor, positive, negative) with same/different class labels. Inputs are PCA-reduced to 64 dimensions and unit-normalized before being passed through the network.

Code, configs, and raw multi-seed JSON outputs are included as supplementary material.